

# JEAN-GABRIEL YOUNG

Assistant Professor

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RESEARCH INTERESTS: Computational Statistics, Complex Systems, Forecasting, Epidemiology

## PROFESSIONAL EXPERIENCE

|                                                                                                                 |           |
|-----------------------------------------------------------------------------------------------------------------|-----------|
| • <i>Agora Intelligence</i> , Data Scientist                                                                    | 2025–2026 |
| • <i>Complex Data LLC</i> , Consulting                                                                          | 2024–     |
| • <i>University of Vermont</i> , Assistant Professor, <a href="#">Department of Mathematics and Statistics*</a> | 2021–     |
| • <i>Université Laval</i> , Professeur Associé <a href="#">Département de Physique</a>                          | 2020–     |
| • <i>University of Vermont</i> , Research Assistant Professor, <a href="#">Department of Computer Science</a>   | 2020–2021 |
| • <i>University of Michigan</i> , Postdoctoral Fellow, <a href="#">Center for the Study of Complex Systems</a>  | 2018–2020 |

## EDUCATION

|                                                                                                          |           |
|----------------------------------------------------------------------------------------------------------|-----------|
| <i>Ph.D. in Physics</i> , Université Laval                                                               | 2014–2018 |
| • <i>Thesis title</i> : Inférence et réseaux complexes <sup>†</sup>                                      |           |
| • <i>Advisors</i> : Louis J. Dubé and Patrick Desrosiers                                                 |           |
| <i>M.Sc. in Physics</i> , Université Laval                                                               | 2012–2014 |
| • <i>Thesis title</i> : De la détection de la structure communautaire des réseaux complexes <sup>‡</sup> |           |
| • <i>Advisors</i> : Louis J. Dubé                                                                        |           |
| <i>B.Sc. in Physics</i> , Theoretical Physics major, Université Laval                                    | 2009–2012 |

## SCHOLARSHIPS, GRANTS AND AWARDS

### Grants

|                                                                                                                          |      |
|--------------------------------------------------------------------------------------------------------------------------|------|
| • <a href="#">EPSCoR Pilot Award</a> . “A Data-Driven Framework for Prediction Market Aggregation” (\$38,300, PI)        | 2025 |
| • <a href="#">NSF</a> . “Altruistic stress, economic networks, and endogenous organizational change” (\$399,653, co-PI)  | 2024 |
| • <a href="#">NSF</a> . “Contagion on Complex Social Systems Conference” (\$47,838, PI)                                  | 2023 |
| • <i>James Jeffords Grant</i> , <a href="#">University of Vermont</a> . “Vermont Open Source Connector” (\$4,600, PI)    | 2023 |
| • <i>OVPR Express Grant</i> , <a href="#">University of Vermont</a> . “Choice Theory in Networks Workshop” (\$3,000, PI) | 2021 |
| • <i>YRNCS Bridge Grant</i> , <a href="#">YRCSS</a> . “Simplicial Configuration Models” (€1,000, PI)                     | 2016 |

### Fellowships and Scholarships

|                                                                                                                           |      |
|---------------------------------------------------------------------------------------------------------------------------|------|
| • <i>Postdoctoral Fellowship in Studying Complex Systems</i> , <a href="#">James S. MacDonnell Foundation</a> (\$200,000) | 2017 |
| • <i>Doctoral Research Scholarship</i> , <a href="#">Fonds de recherche du Québec – Nature et Technologies</a> (\$60,000) | 2014 |

\*Secondary appointment with: Vermont Complex Systems Institute, Larner College of Medicine, Department of Computer Science

<sup>†</sup>Inference and complex networks

<sup>‡</sup>Of community structure detection on complex networks

## Awards

- *FOSS Award*, 2021 Mining Software Repositories Conference 2021
- *Zachary Karate Club Club* award 2021
- *Best oral presentation award*, NERCCS 2020 2020
- *Board of Honour* (Highest overall mark award by all committee members), Ph.D thesis, Université Laval 2018

## PUBLICATIONS

### Peer-reviewed journals

41. [Inferring signed social networks from contact patterns](#) 2026  
D. Ferenczi, **J.-G. Young**, and L. Peel  
J. Phys. Complex. 7, 025018
40. [Sensitivity analysis of epidemic forecasting and spreading on networks with probability generating functions](#) 2026  
M. Boudreau, W. H. W. Thompson, C. Danforth, **J.-G. Young**, and L. Hébert-Dufresne  
J. R. Soc. Interface 23, 20250686
39. [One pathogen does not an epidemic make: A review of interacting contagions, diseases, beliefs and stories](#) 2025  
L. Hébert-Dufresne, Y.-Y. Ahn, A. Allard, V. Colizza, J. W. Crothers, P. Sheridan Dodds, M. Galesic, F. Ghanbarnejad, D. Gravel, R. A. Hammond, K. Lerman, J. Lovato, J. J. Openshaw, S. Redner, S. V. Scarpino, G. St-Onge, T. R. Tangherlini, and **J.-G. Young**  
npj Complexity 2, 26
38. [Symmetry-driven embedding of networks in hyperbolic space](#) 2025  
A. Lizotte, **J.-G. Young**, and A. Allard  
Commun. Phys. 8, 199
37. [Governance as a complex, networked, democratic, satisfiability problem](#) 2025  
L. Hébert-Dufresne, N. W. Landry, J. Lovato, J. St-Onge, **J.-G. Young**, M.-E. Couture-Ménard, S. Bernatchez, C. Choquette, and A. A. Cohen  
npj Complexity 2, 14
36. [Reconstructing networks from simple and complex contagions](#) 2024  
N. W. Landry, W. Thompson, L. Hébert-Dufresne, and **J.-G. Young**  
Phys. Rev. E 110, L042301
35. [Network compression with configuration models and the minimum description length](#) 2024  
L. Hébert-Dufresne, **J.-G. Young**, A. Daniels, A. Kirkley, and A. Allard  
Phys. Rev. E 110, 034305
34. [The simpliciality of higher-order networks](#) 2024  
N. W. Landry, **J.-G. Young**, and N. Eikmeier  
EPJ Data Sci. 13, 17
33. [Hypergraph reconstruction from uncertain data](#) 2023  
A. Lizotte, **J.-G. Young**, and A. Allard  
Sci. Rep. 13, 21364
32. [Accurately summarizing an outbreak using epidemiological models takes time](#) 2023  
B. K. M. Case, **J.-G. Young**, and L. Hébert-Dufresne  
R. Soc. Open Sci. 10, 230634
31. [Opposing responses to scarcity emerge from functionally unique sociality drivers](#) 2023  
A. B. Kao, A. K. Hund, F. P. Santos, **J.-G. Young**, D. Bhat, J. Garland, R. A. Oomen, and H. F. McCreery  
Am. Nat. 202, 3
30. [Exact and rapid linear clustering of networks with dynamic programming](#) 2023  
A. Patania, A. Allard, and **J.-G. Young**  
Proc. R. Soc. A 479, 2275
29. [Compressing network populations with modal networks reveals structural diversity](#) 2023  
A. Kirkley, A. Rojas, M. Rosvall, and **J.-G. Young**  
Commun. Phys. 6, 148

28. [Latent network models to account for noisy, multiply-reported social network data](#) 2023  
C. De Bacco, M. Contisciani, J. Cardoso-Silva, H. Safdari, D. Theuerkauf, T. Sweet, **J.-G. Young**, J. Koster, C. Ross, R. McElreath, D. Redhead, and E. A. Power  
J. R. Stat. Soc. A 186, 355–375
27. [Spatial epidemiology and adaptive targeted sampling to manage the Chagas disease vector \*Triatoma dimidiata\*](#) 2022  
B. K. M. Case, **J.-G. Young**, D. Penados, L. Hébert-Dufresne, and L. Stevens  
PLOS Negl. Trop. Dis. 16, e0010436
26. [Impact and dynamics of hate and counter speech online](#) 2022  
J. Garland, K. Ghazi-Zahedi, **J.-G. Young**, L. Hébert-Dufresne, and M. Galesic  
EPJ Data Sci. 11, 3
25. [Clustering of heterogeneous populations of networks](#) 2022  
**J.-G. Young**, A. Kirkley, and M. E. J. Newman  
Phys. Rev. E 105, 014312
24. [Reconstruction of plant–pollinator networks from observational data](#) 2021  
**J.-G. Young**, F. S. Valdovinos, and M. E. J. Newman  
Nat. Commun. 12, 3911
23. [Hypergraph reconstruction from network data<sup>§</sup>](#) 2021  
**J.-G. Young**, G. Petri, and T. P. Peixoto  
Commun. Phys. 4, 135
22. [A clarified typology of core-periphery structure in networks](#) 2021  
R. J. Gallagher, **J.-G. Young**, and B. Foucault Welles  
Sci. Adv. 7, eabc9800
21. [Bayesian inference of network structure from unreliable data](#) 2021  
**J.-G. Young**, G. T. Cantwell, and M. E. J. Newman  
J. Complex. Netw. 8, cnaa046
20. [Inference, model selection, and the combinatorics for growing trees](#) 2021  
G. T. Cantwell, G. St-Onge, and **J.-G. Young**  
Phys. Rev. Lett. 126, 038301
19. [Networks beyond pairwise interactions: structure and dynamics \(review\)](#) 2020  
F. Battiston, G. Cencetti, I. Iacopini, V. Latora, M. Lucas, A. Patania, **J.-G. Young**, and G. Petri  
Phys. Rep. 874
18. [Improved mutual information measure for classification and community detection](#) 2020  
M. E. J. Newman, G. T. Cantwell, and **J.-G. Young**  
Phys. Rev. E 101, 042304
17. [Macroscopic patterns of interacting contagions are indistinguishable from social reinforcement](#) 2020  
L. Hébert-Dufresne, S. V. Scarpino, and **J.-G. Young**  
Nat. Phys. 16, 426
16. [Phase transition in the recoverability of network history](#) 2019  
**J.-G. Young**, G. St-Onge, E. Laurence, C. Murphy, L. Hébert-Dufresne, and P. Desrosiers  
Phys. Rev. X 9, 041056
15. [Efficient sampling of spreading processes on complex networks using a composition and rejection algorithm](#) 2019  
G. St-Onge, **J.-G. Young**, L. Hébert-Dufresne, and L. J. Dubé  
Comput. Phys. Commun. 240, 30
14. [Universality of the stochastic block model](#) 2018  
**J.-G. Young**, G. St-Onge, P. Desrosiers, and L.J.Dubé  
Phys. Rev. E 98, 032309
13. [Exact analytical solution of irreversible binary dynamics on networks](#) 2018  
E. Laurence, **J.-G. Young**, S. Melnik, and L.J.Dubé  
Phys. Rev. E 97, 032302
12. [Phase transition of the susceptible-infected-susceptible dynamics on time-varying configuration model networks](#) 2018

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<sup>§</sup>Appears in the Focus Collection on Higher-order Interaction Networks

- G. St-Onge, **J.-G. Young**, E. Laurence, C. Murphy, and L. J. Dubé  
Phys. Rev. E 97, 022305
11. [Construction of and efficient sampling from the simplicial configuration model](#) 2017  
**J.-G. Young**, G. Petri, F. Vaccarino, and A. Patania  
Phys. Rev. E 96, 032312
  10. [Strategic tradeoffs in competitor dynamics on adaptive networks](#) 2017  
L. Hébert-Dufresne, A. Allard, P.-A. Noël, **J.-G. Young**, , and E. Libby  
Sci. Rep. 7, 7576
  9. [Finite size analysis of the detectability limit of the stochastic block model](#) 2017  
**J.-G. Young**, P. Desrosiers, L. Hébert-Dufresne, E. Laurence, and L. J. Dubé  
Phys. Rev. E 95, 062304
  8. [Growing networks of overlapping communities with internal structure](#) 2016  
**J.-G. Young**, L. Hébert-Dufresne, A. Allard, and L. J. Dubé  
Phys. Rev. E 94, 022317
  7. [Constrained growth of complex scale-independent systems](#) <sup>¶</sup> 2016  
L. Hébert-Dufresne, A. Allard, **J.-G. Young**, and L. J. Dubé  
Phys. Rev. E 93, 032304
  6. [Complex networks as an emerging property of hierarchical preferential attachment](#) 2015  
L. Hébert-Dufresne, E. Laurence, A. Allard, **J.-G. Young**, and L. J. Dubé  
Phys. Rev. E 92, 062809
  5. [General and exact approach to percolation on random graphs](#) 2015  
A. Allard, L. Hébert-Dufresne, **J.-G. Young**, and L. J. Dubé  
Phys. Rev. E 92, 062807
  4. [A shadowing problem in the detection of overlapping communities](#) 2015  
**J.-G. Young**, A. Allard, L. Hébert-Dufresne, and L. J. Dubé  
PLOS ONE 10, e0140133
  3. [Coexistence of phases and the observability of random graphs](#) <sup>¶</sup> 2014  
A. Allard, L. Hébert-Dufresne, **J.-G. Young**, and L. J. Dubé  
Phys. Rev. E 89, 022801
  2. [Percolation on random networks with arbitrary  \$k\$ -core structure](#) 2013  
L. Hébert-Dufresne, A. Allard, **J.-G. Young**, and L. J. Dubé  
Phys. Rev. E 88, 062820
  1. [Global efficiency of local immunization on complex networks](#) 2013  
L. Hébert-Dufresne, A. Allard, **J.-G. Young**, and L. J. Dubé  
Sci. Rep. 3, 2171

#### Peer-reviewed conference proceeding

5. [Cutting through the noise to infer autonomous system topology](#) 2022  
K. G. Leyba, J. J. Daymude, **J.-G. Young**, M. E. J. Newman, J. Rexford, and S. Forrest  
INFOCOM 2022, Proceedings of the 2022 IEEE International Conference on Computer Communications, pp. 1609–1618.
4. [The OCEAN mailing list data set: Network analysis spanning mailing lists and code repositories](#) 2022  
M. Warrick, S. F. Rosenblatt, **J.-G. Young**, L. Hébert-Dufresne, and J. P. Bagrow  
MSR 2022, Proceedings of the 19th International Conference on Mining Software Repositories
3. [Which contributions count? Analysis of attribution in open source](#) 2021  
**J.-G. Young**, A. Casari, K. McLaughlin, M. Z. Trujillo, L. Hébert-Dufresne, and J. P. Bagrow  
MSR 2021, Proceedings of the 18th International Conference on Mining Software Repositories
2. [Countering hate on social media: Large scale classification of hate and counter speech](#) 2020  
J. Garland, K. Ghazi-Zahedi, **J.-G. Young**, L. Hébert-Dufresne, and M. Galesic  
ACL 2020, Proceedings of the Fourth Workshop on Online Abuse and Harms, pp. 102–112.

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<sup>¶</sup>Editors' suggestion

1. [Connected graphs with a given degree sequence: Efficient sampling, correlations, community detection and robustness](#) 2020  
J. Ring IV, **J.-G. Young**, and L. Hébert-Dufresne.  
NetSci-X 2020, Proceedings of NetSci-X 2020: Sixth International Winter School and Conference on Network Science, pp. 33–47.

#### Other edited works

2. [Book review: Advances in Network Clustering and Blockmodeling](#) 2022  
**J.-G. Young**  
J. Soc. Struct. 23, 47
1. [Open source ecosystems need equitable credit across contributions](#) 2021  
A. Casari, K. McLaughlin, M. Z. Trujillo, **J.-G. Young**, J. P. Bagrow, and L. Hébert-Dufresne  
Nat. Comput. Sci. 1, 2

#### Preprints (7)

- [The network epidemiology of an Ebola epidemic](#)  
L. Hébert-Dufresne, **J.-G. Young**, J. Bedson, L. Skrip, D. Pedi, M. F. Jalloh, B. Raulier, O. Lapointe-Gagné, A. Jambai, A. Allard, and B. Althouse  
arXiv:2111.08686
- [The promise of cross-species coexpression analysis in studying the coevolution and ecology of host-symbiont interactions.](#)  
A. Hund, P. Tiffin, **J.-G. Young**, and D. Bolnick  
arXiv:2206.12711  
In revision, *Evolution*
- [Five misunderstandings in animal social network analysis](#)  
D. Redhead, B. Kawam, **J.-G. Young**, D. Franks, C. S Philson, M. van Duijn, J. Hart, M. B. McElreath, R. McElreath, E. A. Power, C. Ross, S. Sosa, C. Steglich, M. Weiss, and L. J. N. Brent  
Submitted, *Nat. Methods*
- [Message passing for epidemiological interventions on networks with loops](#)  
E. Weis, L. Hébert-Dufresne, and **J.-G. Young**  
arXiv:2509.21596  
Submitted, *Phys. Rev. E*
- [Simpson's paradox explains the ubiquity of nonlinear, threshold, and complex contagions](#)  
L. Hébert-Dufresne, A. Allard, **J.-G. Young**, W. H. W. Thompson, and G. St-Onge  
arXiv:2605.00791
- [The nestedness of higher-order networks](#)  
T. LaRock, Y. Zhang, **J.-G. Young**, N. Eikmeier, R. Lambiotte, and N. W. Landry  
arXiv:2605.18420
- [Emergent contagion complexity: Disentangling mechanistic complexity from correlated heterogeneity](#)  
K. Tang, D. Kaiser, W. H. W. Thompson, **J.-G. Young**, L. Hébert-Dufresne, and N. W. Landry  
arXiv:2607.25139  
Submitted, *Phys. Rev. Lett.*

## TEACHING AND MENTORING

#### Instructor

- STAT-6300: *Bayesian Statistics* F2021, F2022, F2023, F2024
- STAT-6990: *Statistical Network Analysis* S2022, S023
- CS-3993: *Independent Study: Machine Learning with graphs* F2023
- STAT-2510: *Applied Probability* F2024
- CSYS-6993: *Independent Study: Information, Physics, and Computation* F2024

#### Schools and guest lectures

- [CSYS/CS 302: Modeling Complex Systems](#), University of Vermont, Burlington VT, 2020, 2021

- [CNWW](#): Complex Networks Winter Workshop, Québec, Canada 2020, 2023
- [CRM Summer School](#): Spectral Theory and Applications, Québec, Canada 2016

## Supervision

- Postdoctoral fellows:
  - ◊ Leah Keating, University of Vermont 2025–
  - ◊ Nicholas W. Landry, University of Vermont 2022–2024
- Ph.D. students:
  - ◊ William H. Thompson, University of Vermont 2024–
  - ◊ Ariane Lizotte, Université Laval (co-direction with Antoine Allard) 2022–
  - ◊ Nicholas J. Robert, University of Vermont 2021–
  - ◊ B. K. M. Case, University of Vermont 2021–2023
- Master's students:
  - ◊ Aviral Chawla, University of Vermont 2022–2024
  - ◊ Erik Weis, University of Vermont 2021–2023
  - ◊ Ariane Lizotte, Université Laval (co-direction with Antoine Allard) 2020–2022
- Undergraduate students:
  - ◊ James Lemahieu (Honors Thesis), University of Vermont AY 25/26
  - ◊ Erik Arnold (Honors Thesis), University of Vermont AY 25/26
  - ◊ Erin Silver (Research Intern), University of Vermont Summer 2024
  - ◊ Nathan Blanchard (Honors Thesis), University of Vermont AY 24/25
  - ◊ Trevor Blanchard (Honors Thesis), University of Vermont AY 22/23

## INVITED TALKS AND SELECTED CONFERENCE CONTRIBUTIONS

- *"Ranking with Networks: Models, Leadership, and AI Preferences"* 2026  
University of Virginia School of Data Science, Charlottesville, VA, USA (talk)
- *"Designing interventions with message passing on clustered graphs"* 2025  
[Montréal Network Science Workshop 2025](#), Montréal, QC, Canada (invited keynote)
- *"Contagion, models and control."* 2025  
Columbia University, New York, NY (invited lecture)
- *"Message passing for intervention design in networks."* 2024  
[QMDN24](#), Los Alamos, NM (invited talk)
- *"Bayesian framework for inference on heterogeneous waste-water networks."* 2024  
[NetSci 2024](#), Québec, Canada (contributed talk)
- *"Complex or simple? Determining a contagion's type from observational data."* 2024  
[WDPCN24](#), São Paulo, Brazil (invited talk)
- *"What can we learn from low-dimensional representations of networks?"* 2024
  - ▷ [NetSI](#), Boston MA, USA (invited seminar)
  - ▷ [Interaction Data Lab](#), Paris, France (invited seminar)
- *"Modeling the Spread of Clostridioides Difficile in Hospitals"* 2023  
[SIAM DS23](#), Portland, OR, USA (talk)
- *"Quantifying Contagion Complexity"* 2023  
[Dynamics of Interacting Contagions – Santa Fe Institute](#), NM, USA (talk)
- *"Statistical Modeling and Inference for Higher-Order Network Science"* 2023  
[KAIS-Vermont Workshop](#), Seoul, Korea (invited talk)
- *"Uncertain Network Science"* 2021–2023
  - ▷ [Channing Network Science Seminar](#), Boston MA, USA (invited seminar)
  - ▷ [NERCCS 2022 conference](#), Buffalo, NY, USA (invited plenary)
  - ▷ University of Vermont — Mechanical Engineering Seminars, Burlington VT, USA (invited seminar)

- ▷ Central European University–Department of Network and Data Science, online (invited seminar)
  - ▷ CNRS, Centre d'Écologie Fonctionnelle et Évolutive, Montpellier, France (invited talk)
  - ▷ University of Maastricht – Department of Data Analytics and Digitalisation, Maastricht, Netherlands (invited seminar)
- “Which contributions count? Analysis of attribution in open source” 2021–2022
  - ▷ MSR2021, online (talk)
  - ▷ BTV Data Science Meet-up, Burlington, VT, USA (talk)
- “Inference with growing networks” 2021
  - CNWW2020, online (invited talk)
- “Bayesian approaches to network epidemiology” 2020
  - HONS 2020, online (invited talk)
- “Paper Unwind: Network archaeology” 2020
  - School of the NERCCS 2020 conference, Buffalo, NY, USA (invited talk)
- “Efficient and fully bayesian inference of complex networks from noisy data” 2019–2020
  - ▷ Indiana University — CNETS, Bloomington, IN, USA (invited seminar)
  - ▷ Université Laval — CIMMUL, Québec, QC, Canada (invited seminar)
  - ▷ Netsci-X 2020, Tokyo, Japan (talk)
  - ▷ NERCCS 2020, Buffalo, NY, USA (talk, best presentation award)
  - ▷ University of Michigan — Jacobs Lab (UMSI), Ann Arbor MI, USA (invited seminar)
  - ▷ Indiana University — Betzel Lab, Bloomington, IN, USA (invited seminar)
  - ▷ Netsci 2020, online (talk)
- “Compression of treelike complex networks using layered configuration models” 2019
  - Netsci 2019, Burlington, VT, USA (talk)
- “Bayesian inference of effective contagion models from population level data” 2019
  - SINM 2019, Burlington, VT, USA (talk)
- “Universality of the stochastic block model” 2019
  - SYNS Warm-up Event 2019, Burlington, VT, USA (invited talk)
- “The statistical physics of inference for Complex Networks” 2018
  - Department of Physics Colloquium Oakland University, Rochester, MI, USA (invited seminar)
- “Network archaeology: phase transition in the recoverability of network history” 2018
  - ▷ Univeristy of Colorado Boulder — StatOptML seminar, Boulder, CO, USA (invited seminar)
  - ▷ Univeristy of Vermont — Vermont Complex Systems Institute, Burlington, VT, USA (invited seminar)
  - ▷ Netsci 2018, Paris, France (talk)
  - ▷ Sentinel North 2018 Annual Meeting, Québec, Canada (plenary)
  - ▷ Univeristy of Bath — Centre for Networks and Collective Behaviour, Bath, UK (invited seminar)
  - ▷ Connected Past 2018, Oxford, UK (talk)
- “Construction of and efficient sampling from the simplicial configuration model” 2017
  - ▷ HONS 2017, Indianapolis, IN, USA (invited talk)
  - ▷ Indiana University — School of Informatics, Bloomington, IN, USA (invited seminar)
  - ▷ University of Michigan — Center for the Study of Complex Systems, Ann Arbor, MI, USA (invited talk)
- “Statistical mechanics of mesoscopic structure extraction” 2017
  - Netsci 2017, Indianapolis, IN, USA (talk)
- “Finite size analysis of the detectability limit of the stochastic block model” 2016
  - ▷ Netsci 2016, Seoul, Korea (lightning talk)
  - ▷ SINM 2016, Seoul, Korea (talk)
  - ▷ ISI Foundation, Torino, Italy (invited seminar)
- “Structural preferential attachment: scale-free benchmark for overlapping community detection algorithms” 2015
  - Netsci 2015, Zaragoza, Spain (poster)
- “Structural preferential attachment of community structure and its relation to Dunbar’s number” 2014
  - Netsci 2014, Berkeley, CA, USA (talk)
- “Complex networks are an emerging property of hierarchical preferential attachment”<sup>11</sup> 2014
  - NetSci 2014 Science, Berkeley, CA, USA (poster)
- “Local and global solutions to community detection: when resolution matters” 2013
  - NetSci 2013, Copenhagen, Denmark (poster)

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<sup>11</sup> Outstanding poster award

## LEADERSHIP AND SERVICE

### Organizer

- Organizer, [Montréal Network Science Workshop 2026](#), CIRRELT, Montréal, Canada 2026
- Organizer, [CNWW](#), Complex Networks Winter Workshop, Québec, Canada 2021, 2023, 2025
- Program Chair, [NetSci 2024](#) (School and Conference on Network Science) 2024
- Chair, [CCSS23](#) (Contagion on Complex Social Systems) 2023
- Organizer, [SINM](#) (Statistical Inference for Network Models) 2021, 2022, 2023
- Satellite location organizer (UVM), [NERCCS 2022](#) 2022
- Organizer, [SIAM DS 21 Mini-Symposium on Dynamics in Higher-Order Networks](#), online 2021
- Co-chair, First [OpenNetSci Hackathon](#), Burlington VT, USA 2019
- Organizer, [NetSci 2019](#), Burlington VT, USA 2019

### Service

- Contributor, [Several open-source projects](#) ongoing
- Member, Faculty Search Committees, Statistics, UVM 2022, 2023, 2024
- Member, Program Committee, Complex Systems, UVM 2024–
- Seminar chair, [STAT@UVM](#) 2022–
- Seminar chair, [Vermont Complex Systems Institute](#) 2021–2023

### Reviewer

- *Journals* (35): [Science Advances](#), [Nature Communications](#), [SIAM Review](#), [Physical Review X](#), [Physical Review Letters](#), [Psychological Methods](#), [PLOS Computational Biology](#), [JMIR Public Health Surveillance](#), [The Annals of Applied Statistics](#), [Physical Review E](#), [Physical Review Research](#), [EPJ Data Science](#), [Scientific Data](#), [Cambridge Elements](#), [EPL](#), [Journal of Open Source Software](#), [Journal of Physics: Complexity](#), [Journal of Physics A](#), [Journal of Applied and Computational Topology](#), [npj Complexity](#), [Scientific Reports](#), [PLOS Complex Systems](#), [PLOS ONE](#), [Palgrave Communications](#), [Journal of Complex Networks](#), [Physics Letter A](#), [Chaos Solitons & Fractals](#), [Entropy](#), [Network Science](#), [Animal Behaviour](#), [Applied Network Science](#), [Statistics and Computing](#), [Knowledge and Information Systems](#), [Journal of Computational Science](#), [Chaos](#).
- *Grants*: Panelist, NSF, IIS Division (2019).

### Program committee

- Northeast Regional Conference on Complex Systems (NERCCS) 2020, 2021, 2022, 2024, 2025
- International School and Conference on Network Science (NetSci) 2019, 2020, 2023, 2024(X), 2025, 2026
- International Conference on Complex Networks and their Applications 2023
- SIAM Workshop on Network Science (SIAM NS) 2018, 2020

### PhD thesis committees

- [Tung-Lin Liu](#), Food Systems. Advisor: Christopher Koliba ongoing
- [Lucy Greenberg](#), Statistics. Advisor: Jeffrey S. Buzas ongoing
- [Larry D. Long](#), Complex Systems and Data Science. Advisor: Britt Williams 2025
- [Nicolò Ruggeri](#) (ETH), Machine Learning and Network Science. Advisor: Caterina de Bacco 2024
- [Mariah Bourdreau](#), Mathematics. Advisor: Laurent Hébert-Dufresne 2024
- [Samuel Rosenblatt](#), Computer Science. Advisor: Laurent Hébert-Dufresne 2024
- [Damin Zhu](#), Statistics. Advisor: Jeffrey S. Buzas 2023
- [Michael Arnolds](#), Complex Systems. Advisor: Peter Dodds 2023

## SELECTED SOFTWARE

(Complete list available online)

- bayesian-workflow (claude-plugin)
- Bayesian inference of networks from noisy data (stan)
- Bayesian inference of effective contagion models from population level data (stan)
- Reconstruction of plant-pollinator networks from observational data (stan + python)
- Sequential MC sampler for Network Archaeology (python + C++)
- MCMC sampler for the Simplicial Configuration Model (C++)
- MCMC sampler for the Stochastic Block Model (C++)
- Structural Preferential Attachment community detection benchmark (C++)

## VARIA

### Selected media coverage

- |                                                                                                             |      |
|-------------------------------------------------------------------------------------------------------------|------|
| • “Au temps de la Révolution française, des rumeurs propagées comme des virus.” Les années lumière (French) | 2025 |
| • “Physiology-inspired networks could improve political decision-making.” Phys.org                          | 2025 |
| • “Are ideas contagious?.” Phys.org                                                                         | 2024 |
| • “A selection of 2020’s highlighted research.” Nature                                                      | 2021 |
| • “To find the right network model, compare all possible histories.” Phys.org                               | 2021 |
| • “Fighting Hate Speech with AI & Social Science,” Complexity Podcast                                       | 2020 |
| • “How you talk about coronavirus actually impacts its spread,” cnet                                        | 2020 |
| • “Neue Studie zeigt Wirksamkeit von Gegenrede im Netz,” netzpolitik.org                                    | 2020 |
| • “When coronavirus is not alone,” Phys.org                                                                 | 2020 |
| • “The shape of randomness.” Physics Central                                                                | 2017 |
| • “What algae can tell us about political strategy.” Phys.org                                               | 2017 |
| • “L’univers complexe de Jean-Gabriel Young.” Le Soleil (French)                                            | 2017 |